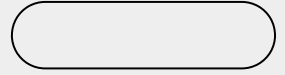


Prototyping Deck



Spring '26

TEAC 882B

Advanced Web Design and Databases

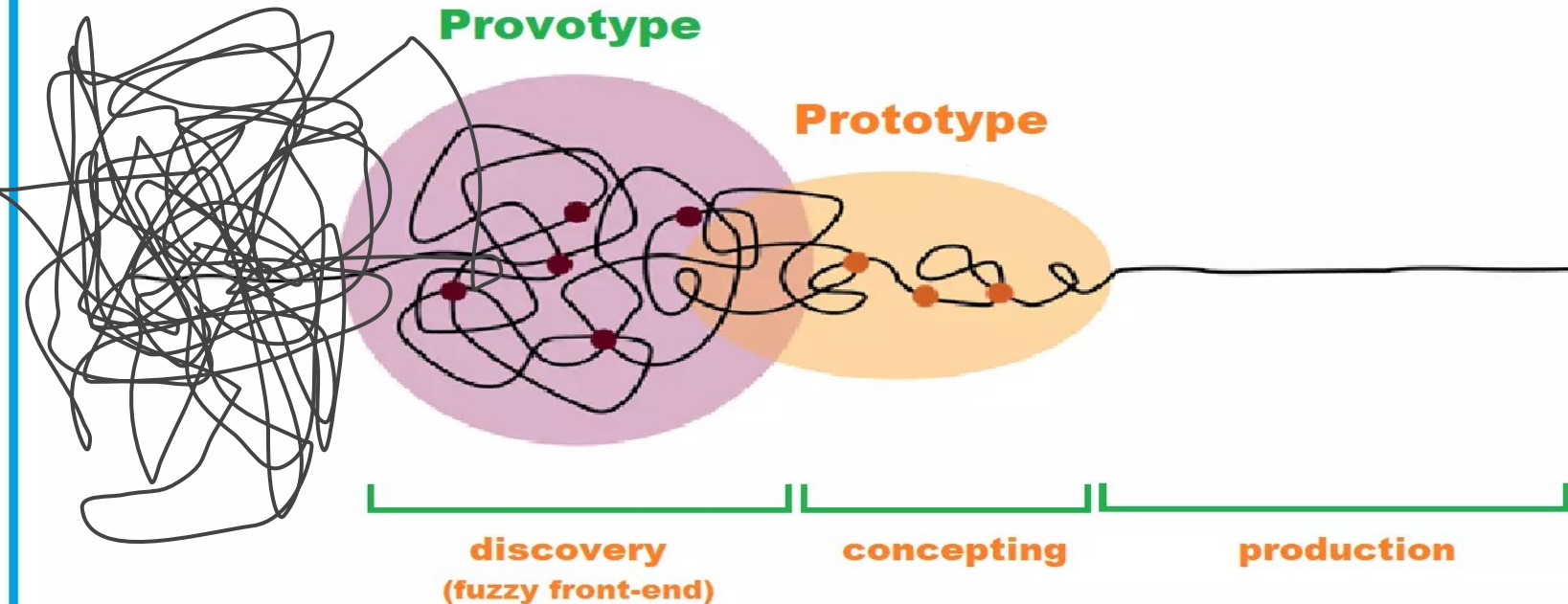
“

I don't know if you can be a good innovator if you don't know how to make the stuff you are designing.

-Walter Isaacson 2023

What is prototyping?

➡ **Conversion of an intangible idea either related to physical or digital world into a tangible configuration to test its potential.**



Personal / Professional Areas of Interest

List your technology skills and any additional interests.

Gaming (Analog and Digital)

Crocheting

Video Editing



Servo

Brainstorming / Ideation Slide

Bucket o'ideas

1. Pain Tracker App (Convert paper template)
2. Sudoku App (Pencil input, solve a math problem for a hint)
3. Flesh out/get working Educational Lens webapp from last semester
4. Chatbot for self interview protocol for dissertation

Second bucket o'ideas

1. Forensic server for SCS 275
2. iOS app version of Operation Neptune
3. Chatbot for self interview protocol for dissertation
4. Habit tracker
5. Student Agency Math Problem Generator (AI-powered customization)
6. Dissertation Autoethnography Tracker (reflection through making)

Servo - Two Interconnected Prototypes



1. Student Agency Math Problem Generator

AI-powered tool where students choose topic, difficulty, and real-world context (sports, music, gaming, cooking, etc.).

System generates personalized math problems. Empowers students who have been underserved by traditional math instruction to direct their own learning.

2. Dissertation Autoethnography Tracker

Personal web app for documenting formative memories with tech/math/education, then logging the process of building the Problem Generator as a through-line for reflection.

AI-assisted prompts connect what I'm building to my pedagogical history. References past conversations for deeper synthesis. Feeds directly into dissertation work.

Servo - Two Interconnected Prototypes



Student Agency Math Problem Generator

What is it?

AI-powered web app where students select math topic, difficulty level, and personal context for problem generation.

Why does it matter?

Empowers students who have been underserved by traditional instruction through agency and personal relevance. Builds student ownership of their math learning.

What's the biggest risk?

AI-generated problems might lack pedagogical quality or real-world authenticity. Need to test with actual students to refine.

What is the closest analogue?

Khan Academy's practice problems, but with student customization and AI generation.

Dissertation Autoethnography Tracker

What is it?

Personal tracker for formative memories with tech/math/education, then logging the Generator build process as reflection through making. AI-assisted reflection connects past to present.

Why does it matter?

Feeds directly into dissertation work. Creates autoethnographic data while the prototyping process demonstrates my evolving pedagogical thinking in practice.

What's the biggest risk?

Might not use it consistently. Need simple, frictionless logging to maintain the habit throughout the semester.

What is the closest analogue?

Research journal meets development log, with AI helping spot patterns between past experiences and current decisions.

I'm selecting these ideas because: [X] it solves a clear problem; [X] I can prototype it sooner; [] I'm willing to abandon it if it fails fast; [X] I'm curious about how it will turn out; [X] It will push me to grow without being impossibly hard.

Servo Student Agency Math Problem Generator

Core Problem / Opportunity

Who is this for?

Students underserved by traditional math instruction

What problem are they experiencing?

Lack of agency and personal relevance in math learning

What will the prototype do?

Primary Interaction (what the user does):

Student selects math topic, difficulty level, and personal context (sports, cooking, gaming, etc.)

System response:

AI generates customized math problems based on student parameters

Success looks like:

Students engage with math problems that feel personally relevant to their lives and interests

Technical Components

Frontend:

React interface, simple form inputs for parameter selection

Data: Claude API for problem generation, browser storage for session data

AI role:

Generate contextually appropriate math problems based on student parameters

Build Plan

Phase 1 (first version working):

Basic interface, topic, AI generated problem

Phase 2 (second version):

Multiple difficulty levels, multiple problem types, answer checking

If I get that far (third version capability):

Progress tracking, history, hint system

Servo Dissertation Autoethnography Tracker

Phase 1: Memory mapping sessions

- Capture formative experiences with tech/math/education (both as adult and child)

Phase 2: Active logging interface

- Document decisions, challenges, breakthroughs while building the Problem Generator

Phase 3: AI-assisted reflection prompts

- System connects current work to documented history
- Example: "You mentioned struggling with rigid math curricula as a student. How is that shaping your design choices here?"

Phase 4: Past conversation integration

- Reference our chat history for relevant discussions
- Surface past conversations about pedagogy, equity, dissertation topic

Phase 5: Timeline visualization

- See pedagogical evolution over time
- Identify patterns, contradictions, growth in thinking

Phase 6: Synthesis/Analysis layer

- AI helps identify patterns across entries
- Track how "equity" and "access" meanings evolve
- Artifact integration (screenshots, code snippets, student feedback)

Neo4j Integration: What Was Built

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Neo4j Aura

Graph Database

- › Free tier cloud instance
- › Nodes: TrackerEntry, Theme, Era
- › Relationships wire entries to contexts and themes automatically



Cloudflare Worker

API Proxy

- › Separate Worker from Anthropic proxy
- › Neo4j query/v2 HTTP API
- › Secure — credentials never touch the browser



Tracker Integration

Frontend

- › neo4j-client.js wired into tracker.js
- › Every save → localStorage + Neo4j
- › Health check on page load
- › Fire-and-forget writes

Data Flow:

User saves entry → localStorage (UI) + Neo4j.saveEntry() → Cloudflare Worker → Neo4j Aura /query/v2 → Graph nodes

TrackerEntry

id · type · title · content
context · tags · createdAt

Era

name
(As a Student / Teacher /
Researcher / Personal)

Theme

name
(auto-created from tags)

Relationships

Entry -[:SITUATED_IN]→ Era

Entry -[:SURFACES]→ Theme

Auto-created on every
save — no extra work
required from the UI.

What We Learned Getting Here

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Obstacles Encountered

Neo4j driver won't run in Cloudflare Workers

Workers don't support raw TCP/Bolt connections. Solution: use the Neo4j HTTP API instead — pure fetch, works perfectly.

Wrong API endpoint — /tx/commit returns 403

Aura free tier blocks the legacy transactional API. Discovered the correct endpoint is /db/{database}/query/v2 via live curl testing.

Wrong database name in credentials file

The Aura credentials file listed the instance ID (22fda3fb) as both hostname AND database name. Database name should be 'neo4j' — then 22fda3fb after further testing.

neo4j-client.js had wrong Worker URL + bad payload

Claude Code used a placeholder URL that never got updated, and wrapped the entry in an extra {entry: ...} object the Worker didn't expect.

Git merge conflict on push

Local changes and Claude Code's remote commits diverged. Resolved with git pull --rebase and git checkout --ours to keep the working v2 Worker.



What's Working Now

- ✓ Worker deployed at dissertation-neo4j.math-generator.workers.dev
- ✓ GET /health returns {"neo4j": "connected"}
- ✓ Every tracker save writes a node to Aura in real time
- ✓ Era + Theme relationships auto-created on each write



What's Next

Claude API conversation integration

Typed + spoken conversations with Claude stored directly as graph nodes alongside tracker entries

schema.cypher constraints

Run uniqueness constraints and seed Era nodes in Aura console — database hygiene before real data accumulates

Ingestion pipeline (TBD)

Written artifacts, images, test scores, and web content → Claude API extraction → Neo4j nodes

See It Live: Graph Viewer

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What It Shows

Live data from Neo4j Aura

Hits the Cloudflare Worker on load — every node in the graph is real dissertation data, pulled fresh each visit

Force-directed graph layout

TrackerEntry (teal), Era (red), and Theme (green) nodes float and settle using D3 physics — draggable, zoomable, pannable

Node inspector panel

Click any node to see its full content, metadata, tags, and all its relationships — click connections to hop the graph

Real-time round-trip demo

Save a memory in the tracker → reload the viewer → new node appears in the graph. The full pipeline visible in 10 seconds

Node Types in the Graph



TrackerEntry

Memories · Build Logs · Reflections



Era

As a Student · Teacher · Researcher · Personal



Theme

Auto-created from tags on each entry

Open in your browser →

<https://www.servellon.net/dissertation-tracker/graph-viewer.html>

No login required — publicly accessible, live data

Live Demo Sequence

1

Open tracker → save a memory with tags

3

Open graph-viewer.html → new node visible

2

Console shows: [Neo4j] write succeeded

4

Click node → inspector shows full content